Answer Key — XII CS Preparatory-1

Section A (MCQs & Assertion-Reason)

Q	Answer
1	**b) The finally block can be used to execute statements whether an exception occurs or not.**
2	**d) 30** **Completed**
3	**b) Domain**
4	**b) ALTER**
5	**b) AS**
6	**b) (8,**
7	**c) phone IS NOT NULL**
8	**c) Cardinality**
9	**a) DATE**
10	**b) 20**
11	**b) fetchall()**
12	**c) DESCRIBE student;**
13	**a) Both A and R are True and R is the correct explanation for A.**
14	**a) Both A and R are True and R is the correct explanation for A.**

Section B (2 marks each)

15. A. Candidate Key & Alternate Key

> **Candidate Key:** Any attribute (or minimal set of attributes) that can uniquely identify each tuple in a relation.
> **Alternate Key:** A candidate key that was *not* chosen as the primary key.
> **Relation:** Alternate Key = Candidate Key − Primary Key.

OR

> **B. Degree vs Cardinality**

Degree	Cardinality
Number of attributes (columns) in a relation	Number of tuples (rows) in a relation
Example: In `STUDENT(Roll, Name, Marks)`, degree = 3	If 50 students are stored, cardinality = 50

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16. Python-MySQL Connection Program

```
import mysql.connector
con = mysql.connector.connect(host="localhost", user="root",
                              password="school123", database="schooldb")
if con.is_connected():
    print("Database Connected Successfully")
con.close()
```

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17. Output

```
Cannot divide by zero
End of calculation
```

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18. A.

```
I.  SHOW DATABASES;
II. CREATE DATABASE STANES;
```

OR

B.

```
I.  USE STANES;
II. SHOW TABLES;
```

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19. DELETE vs DROP

DELETE	DROP
Removes rows (tuples) from a table	Removes the entire table (structure + data)
Can use WHERE clause	Cannot use WHERE clause
Table structure remains	Table structure is deleted
Example: `DELETE FROM faculty WHERE faculty_id=101;`	Example: `DROP TABLE faculty;`

20. `insert_faculty()` Function

```
import mysql.connector

def insert_faculty():
    con = mysql.connector.connect(host="localhost", user="root",
                                  password="school123", database="schooldb")
    cur = con.cursor()
    cur.execute("INSERT INTO faculty VALUES (107, 'Anita', 55000)")
    con.commit()
    print("Record inserted")
    con.close()
```

Section C (4 marks each)

21. Queries on EMPLOYEE

```
I.  SELECT Emp_Name, Salary FROM EMPLOYEE WHERE Salary > 60000;
II. SELECT Department, SUM(Salary) FROM EMPLOYEE GROUP BY Department;
III.SELECT * FROM EMPLOYEE WHERE Emp_Name LIKE '%a';
IV. SELECT * FROM EMPLOYEE ORDER BY Salary ASC;
```

22. Update Marks (Python-MySQL)

```
import mysql.connector

con = mysql.connector.connect(host="localhost", user="admin",
                              password="admin@123", database="schooldb")
cur = con.cursor()
cur.execute("UPDATE RESULT SET Marks=72 WHERE Student_ID=204")
con.commit()
print(cur.rowcount, "row(s) updated")
con.close()
```

OR — Delete Record

```
import mysql.connector

con = mysql.connector.connect(host="localhost", user="admin",
                              password="admin@123", database="schooldb")
cur = con.cursor()
cur.execute("DELETE FROM RESULT WHERE Student_ID=204")
con.commit()
print("Rows affected:", cur.rowcount)
con.close()
```

23. Queries on SPORTS_REG

```
I.  SELECT Sport_Name, COUNT(*) FROM SPORTS_REG GROUP BY Sport_Name;
II. SELECT Player_Name FROM SPORTS_REG WHERE Player_Name LIKE '%ri%';
III.SELECT MAX(Reg_Fee) FROM SPORTS_REG;
IV. SELECT Sport_Name, AVG(Reg_Fee) FROM SPORTS_REG GROUP BY Sport_Name;
```

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24. Outputs

I. `SELECT Player_Name, Sport_Name FROM SPORTS_REG WHERE Grade IN (10,11) ORDER BY Reg_ID;`

Player_Name	Sport_Name
Aarav	Cricket
Riya	Football
Navya	Badminton
Rohit	Athletics

II. `SELECT Sport_Name, COUNT(*) FROM SPORTS_REG GROUP BY Sport_Name ORDER BY Sport_Name;`

Sport_Name	COUNT(*)
Athletics	1
Badminton	1
Cricket	2
Football	2

III. `SELECT Player_Name FROM SPORTS_REG WHERE Player_Name LIKE 'R%' ORDER BY Player_Name;`

Player_Name
Riya
Rohit

IV. `SELECT SUM(Reg_Fee) FROM SPORTS_REG WHERE Grade BETWEEN 10 AND 11;`

SUM(Reg_Fee)
2700

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25. Queries on EVENT & REGISTRATION

```
I.  UPDATE EVENT SET Venue='Computer Lab' WHERE Event_Name='Coding Contest';

II. SELECT * FROM EVENT WHERE Event_Date > '2026-02-01';

III.SELECT R.Student_Name
    FROM REGISTRATION R
    JOIN EVENT E ON R.Event_ID = E.Event_ID
    WHERE E.Venue = 'Main Hall';

IV. SELECT * FROM EVENT, REGISTRATION;
```

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26. Python-MySQL Program

```
import mysql.connector

try:
    con = mysql.connector.connect(host="localhost", user="admin",
                                  password="admin@123", database="College")
    cur = con.cursor()
    cur.execute("SELECT * FROM Faculty")
    records = cur.fetchall()
    for row in records:
        print(row)
except mysql.connector.Error as e:
    print("Error:", e)
finally:
    if con.is_connected():
        con.close()
```
